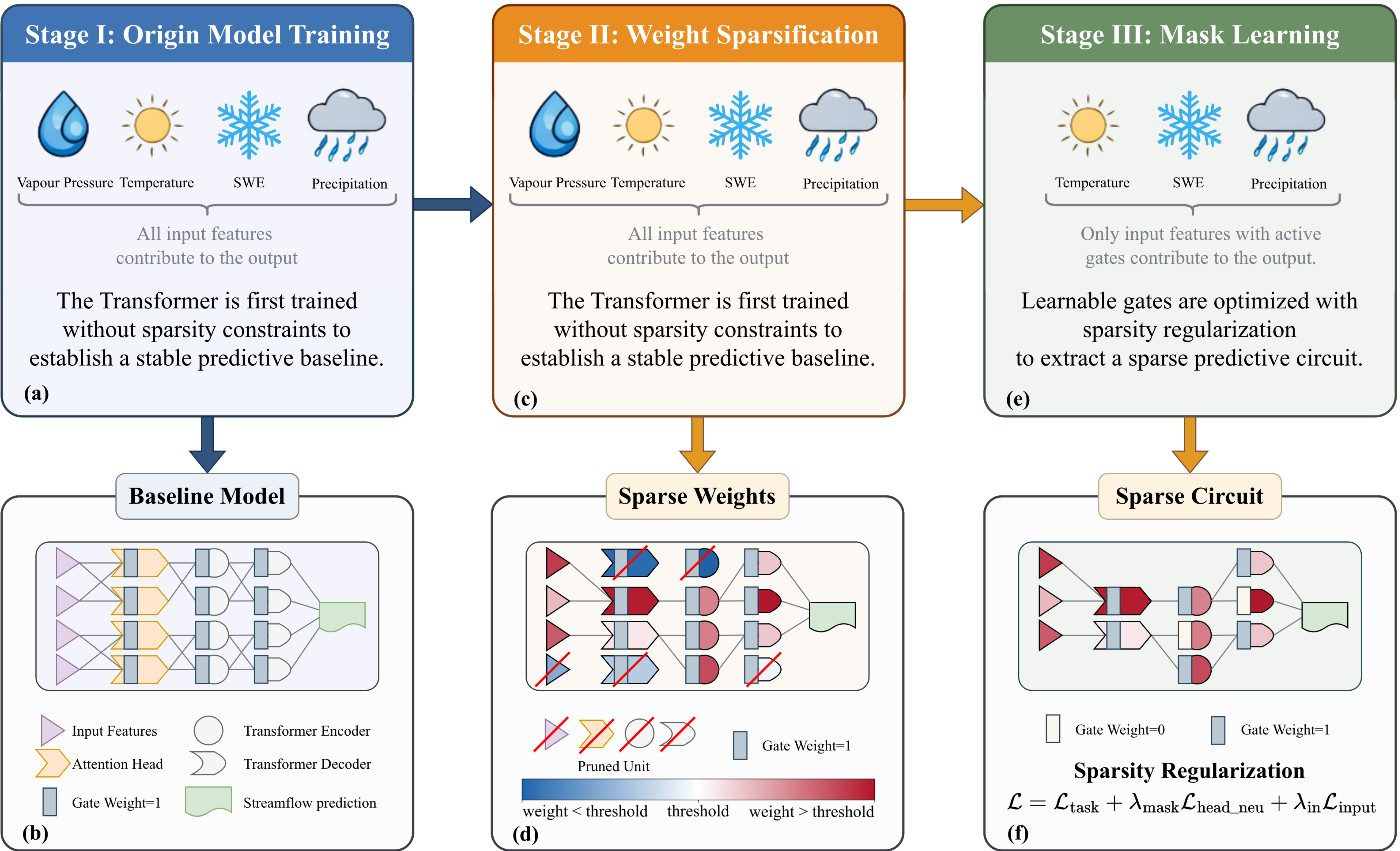

Stage I: Origin Model Training
Vapour Pressure
Temperature
SWE
Precipitation
All input features contribute to the output
The Transformer is first trained without sparsity constraints to establish a stable predictive baseline.
(a)
Stage II: Weight Sparsification
Vapour Pressure
Temperature
SWE
Precipitation
All input features contribute to the output
The Transformer is first trained without sparsity constraints to establish a stable predictive baseline.
(c)
Stage III: Mask Learning
Temperature
SWE
Precipitation
Only input features with active gates contribute to the output.
Learnable gates are optimized with sparsity regularization to extract a sparse predictive circuit.
(e)
Baseline Model
Input Features
Transformer Encoder
Attention Head
Transformer Decoder
Gate Weight=1
Streamflow prediction
(b)
Sparse Weights
Pruned Unit
Gate Weight=1
weight < threshold
threshold
weight > threshold
(d)
Sparse Circuit
Gate Weight=0
Gate Weight=1
Sparsity Regularization
$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{mask}}\mathcal{L}_{\text{head\_neu}} + \lambda_{\text{in}}\mathcal{L}_{\text{input}}$
(f)